

# Front-End Development Viva Questions & Answers (All 14 Units)

## UNIT 1: BOOTSTRAP

1. What is Bootstrap and why is it used?

Bootstrap is a free front-end framework used to create responsive and mobile-first websites quickly using HTML, CSS, and JavaScript.

2. List the advantages of Bootstrap 4.

- Easy to use with basic HTML/CSS knowledge.
- Responsive design for all devices.
- Compatible with all modern browsers.
- Fewer cross-browser bugs.
- Lightweight.
- Strong grid system.

3. How does the Bootstrap Grid System work?

Bootstrap uses a 12-column grid system that allows responsive layout design.

4. What are the container classes in Bootstrap?

- .container: Fixed-width, responsive.
- .container-fluid: Full-width.

5. What is the <meta viewport> tag used for?

It ensures proper scaling and zooming on mobile devices.

6. Types of Bootstrap form layouts?

- Vertical, Horizontal, Inline

7. What is a Bootstrap modal?

A popup dialog box over the current page.

8. What is Carousel in Bootstrap?

Slideshow for cycling through content.

9. .navbar-default vs .navbar-inverse?

Default: Light; Inverse: Dark navbar.

10. What is CDN in Bootstrap?

CDN loads Bootstrap files from remote servers for faster access.

11. Badges and Labels?

Badges: Number indicators; Labels: Info tags like .label-primary

12. Background color classes?

.bg-primary, .bg-success, .bg-warning, etc.

## UNIT 2: GIT

1. What is Git?

Git is a distributed version control system.

2. Git vs GitHub?

Git: Tool; GitHub: Hosting service.

### 3. Basic Git Commands:

git init, git add, git commit -m "msg", git status, git push, git clone

### 4. git commit -m "msg"?

Saves changes with a message.

### 5. Push code to GitHub?

git remote add origin <url>, git push origin main

### 6. What is a repository?

Project folder with version history.

### 7. Who created Git?

Linus Torvalds in 2005.

### 8. Full Stack Developer?

Works on both frontend and backend.

## UNIT 3: NODE.JS & NPM

### 1. What is Node.js?

JavaScript runtime for backend development.

### 2. What is NPM?

Package manager for Node.js.

### 3. Check Node.js?

node -v

### 4. What does NPM do?

Installs, manages dependencies.

### 5. Steps to install Node.js?

Download from [nodejs.org](https://nodejs.org), install, check with node -v

### 6. Is Node.js asynchronous?

Yes.

## UNIT 4: RESPONSIVE DESIGN

### 1. What is responsive design?

Web adapts to all device sizes.

### 2. What are media queries?

CSS techniques for different screen sizes.

### 3. Fixed vs Fluid vs Responsive?

Fixed: px; Fluid: %; Responsive: adjusts via breakpoints.

## UNIT 5: NAVIGATION

### 1. What are navigation bars?

Used for site links, responsive headers.

2. What is `.navbar-expand-*`?

Defines screen size breakpoint.

## UNIT 6: NAVIGABILITY ELEMENTS

1. What is a breadcrumb?

Shows path/location in hierarchy.

2. What is pagination?

Breaks content into pages, uses `.pagination`.

## UNIT 7: BOOTSTRAP CSS

1. What are cards?

Content containers with styling.

2. What are Jumbotron?

Large content boxes (removed in BS5).

## UNIT 8: FORMS

1. What are input groups?

Add-ons to input fields.

2. Form validation?

Use `.is-valid`, `.is-invalid` classes.

## UNIT 9: TABS & PILLS

### 1. What are tabs and pills?

Tab-based navigation using `.nav-tabs/.nav-pills`.

### 2. How to toggle tab content?

Use `data-bs-toggle` and matching `id`.

## UNIT 10: COLLAPSE & MODALS

### 1. What is Collapse?

Expandable/collapsible content.

### 2. Trigger modal?

Use `data-bs-toggle="modal"` with `id`.

## UNIT 11: TOOLTIPS

### 1. How to use tooltips?

Add `data-bs-toggle="tooltip"`, init using JS.

## UNIT 12: JQUERY

### 1. What is jQuery?

JS library for DOM, AJAX, effects.

## 2. jQuery vs JavaScript?

jQuery is a simplified wrapper.

## UNIT 13: SASS

### 1. What is SASS?

CSS preprocessor with variables, mixins.

### 2. SCSS vs SASS?

SCSS uses CSS-like syntax; SASS uses indentation.

## UNIT 14: NPM SCRIPTS

### 1. What are NPM scripts?

Automation commands in package.json.

### 2. Example:

```
"scripts": {  
  "start": "node app.js",  
  "test": "echo Running tests"  
}
```